



QUANTITATIVE METHODS 1 · CHAPTER 4

Bivariate Distributions

Detailed solutions

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1. Reading a relationship

A scatter plot of advertising expenditure and sales slopes upward, with points close to a line except for one distant observation.

- 1 Describe direction, form and strength.
- 2 What should be investigated about the distant point?
- 3 Does the plot prove advertising causes sales?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. The direction is positive, the form is approximately linear and the association appears strong. The distant point may be a data error, an unusual campaign or an influential observation; verify it and compare results with and without it.

Interpretation. The plot does not prove causality because seasonality, product quality or reverse causality may affect both variables.

2. Covariance and correlation

For $x=(1,2,3,4)$ and $y=(2,3,5,6)$ calculate population covariance and correlation.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. $\bar{x}=2.5$ and $\bar{y}=4$. Cross-deviations are 3, .5, .5, 3, summing to 7; squared-deviation sums are 5 for x and 10 for y . Result. $\text{Cov}(x,y)=7/4=1.75$, $\sigma_x=\sqrt{5/4}=1.118$, $\sigma_y=\sqrt{10/4}=1.581$, hence $r=1.75/(1.118 \times 1.581)=.990$.
Interpretation. The variables have a very strong positive linear association; the unit-free correlation is easier to compare than covariance.

3. Least-squares regression

Use the data in Exercise 2 to estimate $y=a+bx$. Forecast y for $x=3.5$ and calculate the residual at (3,5).

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. $b=\text{Cov}(x,y)/V(x)=1.75/1.25=1.4$ and $a= y-b x=4-1.4(2.5)=.5$. Result.

$y=.5+1.4x$. At $x=3.5$, $y=5.4$. At $x=3$, $y=4.7$, so residual $e=y- \hat{y}=5-4.7=.3$.

Interpretation. A one-unit increase in x is associated with an estimated 1.4-unit increase in y within the observed range.

4. R^2 , interpolation and extrapolation

Suppose a model fitted for advertising budgets from EUR 1,000 to EUR 10,000 has $R^2 = .81$. Interpret R^2 . Classify predictions at EUR 7,000 and EUR 20,000 and explain the risk.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. The model explains 81% of the observed variation in sales around their mean; it does not mean that advertising causes 81% of sales. EUR 7,000 is interpolation, while EUR 20,000 is extrapolation. Interpretation. Extrapolation is risky because the linear relationship may not continue beyond the data and market saturation may appear.

5. From association to a decision

A retailer finds $r=.85$ between discounts and weekly sales and recommends permanent large discounts. Give a statistically responsible response including at least four checks.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Check the scatter plot for non-linearity and influential points; inspect residuals; control for holidays, advertising and store differences; examine whether discounting was targeted to weak weeks; distinguish revenue from profit; and, where feasible, use a randomised or credible quasi-experimental design. Interpretation. Strong correlation supports prediction inside the observed setting but is insufficient for a pricing policy because causal effects and margins determine profitability.

6. Complete regression from paired data

For six stores, weekly advertising x (€000) and sales y (€000) are:

x	1	2	3	4	5	6
y	3	5	4	7	8	9

- 1 Draw and describe the scatter plot.
- 2 Given $\bar{x} = 3.5$, $\bar{y} = 6$, $\Sigma(x-\bar{x})^2 = 17.5$ and $\Sigma(x-\bar{x})(y-\bar{y}) = 21$, estimate the regression line.
- 3 Predict sales for $x = 4.5$.
- 4 Calculate the residual for the store with $x = 4$.

Method prompt. Use $b = S_{xy}/S_{xx}$ and $a = \bar{y} - b\bar{x}$. A residual is observed minus fitted.

Detailed solution

Graphical reading. The points show a positive, approximately linear relationship with moderate scatter. No observation should be removed solely from the graph.

Slope. $b = 21/17.5 = 1.2$. Intercept. $a = 6 - 1.2(3.5) = 1.8$. Thus $\hat{y} = 1.8 + 1.2x$.

Prediction. At $x = 4.5$, $\hat{y} = 1.8 + 1.2(4.5) = 7.2$ thousand euros.

Residual. At $x = 4$, the fitted value is 6.6 and the observed value is 7; $e = 7 - 6.6 = +0.4$. The model underpredicts this store by €0.4 thousand.

Interpretation. Within the observed range, an additional €1,000 of advertising is associated with €1,200 higher predicted weekly sales. This is not automatically a causal return on advertising.

7. Regression diagnosis and responsible decision

A model linking store traffic to temperature reports $r = 0.78$ and $R^2 = 0.61$. Its residual plot is U-shaped, one summer weekend has a very large residual, and the manager wants to forecast a temperature 12°C above the observed maximum.

- 1 Interpret r and R^2 correctly.
- 2 Diagnose the residual pattern.
- 3 Discuss the unusual weekend and the proposed forecast.
- 4 Recommend the next analytical steps.

Method prompt. Separate strength, functional form, unusual observations, extrapolation and causality.

Detailed solution

Correlation. $r = 0.78$ indicates a fairly strong positive linear association in the observed sample. It does not establish causality.

R^2 . About 61% of the sample variation in traffic around its mean is explained by the fitted linear relationship. It does not mean that temperature “causes 61%” of traffic.

Residual diagnosis. A U-shape is systematic structure, indicating that a straight line misses curvature. Predictions may be biased at low and high temperatures.

Unusual weekend. Check data quality and contextual factors such as a festival, promotion or closure. Report sensitivity with and without the observation; do not delete it mechanically.

Extrapolation. Forecasting 12°C beyond the maximum is unsupported and particularly unsafe because curvature is already visible.

Next steps. Add a quadratic term or use a flexible curve, validate on held-out periods, include calendar and promotional controls, inspect influence statistics, and present prediction uncertainty.